Vidyavardhini's College of Engineering & Technology

Department of Computer Science and Engineering (Data Science)

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Title:	To perform version control operations using Git for managing source code, tracking changes, and enabling collaborative software development
Date of Performance:	
Date of Submission:	
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Sign of Faculty:



Aim : To perform version control operations using Git for managing source code, tracking changes, and enabling collaborative software development

Theory:

Git Commands

For Setup:

- git config --global user.name "Enter User name of Github Account"
- git config --global user.email "Enter email of Github Account"

For Initialization:

- git init : Initialize an existing Directory as a Git Repository.
- git clone [url] : Retrive an entire repository from a hosted location via URL (Paste HTTPS OR SSH key From your gitHub)

Stage and SnapShot :

(Following commands works with respect to staging area)

- git status : show modified files in working directory, staged for your next commit ● git add [file] : add a file to a staging area
- git reset [file] : (get file back from staging area)unstage a file while retaining the changes in working directory
- git diff : diff of what is changed but not staged
- git diff --staged : diff of what is staged but not yet committed
- git commit -m "[type a message]" : commit your staged content as a new commit snapshot

Branch & merge :

(work in branches, changing context, and integrating changes)

- git branch : list your branches. a * means currently active branch
- git branch [branch-name] : create a new branch at the current commit ● git checkout : switch to another branch and check it out into your working directory ● git merge [branch] : merge the specified branch's history into the current one ● git log : show all commits in the current branch's history

Inspect & compare

● git log : show the commit history for the currently active branch
● git log : branchB..branchA : show the commits on branchA that are not on branchB ● git log --follow [file] : show the commits that changed file, even across renames ● git diff branchB...branchA : show the diff of what is in branchA that is not in branchB ● git show [SHA] : show any object in Git in human-readable format

Share & update :

(Retrieving updates from another repository and updating local repos)

● git remote add [alias] [url] : add a git URL as an alias
● git fetch [alias] : fetch down all the branches from that Git remote
● git merge [alias]/[branch] : merge a remote branch into your current branch to bring it up to date
● git push [alias] [branch] : Transmit local branch commits to the remote repository branch ● git pull : fetch and merge any commits from the tracking remote branch

Tracking path changes :

(Versioning file removes and path changes)

● git rm [file] : delete the file from project and stage the removal for commit ● git mv [existing-path] [new-path] : change an existing file path and stage the move ● git log --stat -M : show all commit logs with indication of any paths that moved

Rewrite history

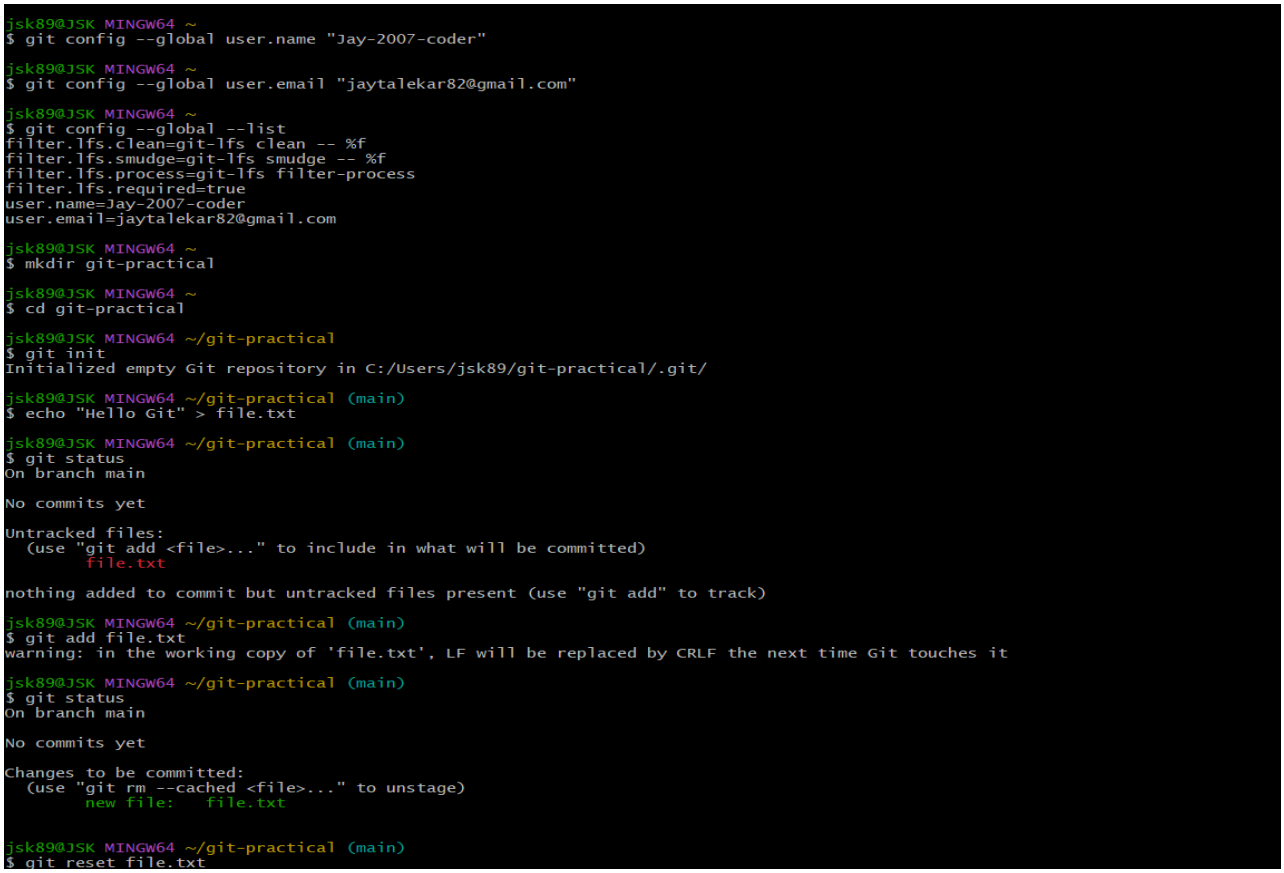
(Rewriting branches, updating commits and clearing history)

● git rebase [branch] : apply any commits of current branch ahead of specified one ● git reset --hard [commit] : clear staging area, rewrite working tree from specified commit ● git stash :

Save modified and staged changes

- git stash list : list stack-order of stashed file changes
- git stash pop : write working from top of stash stack
- git stash drop : discard the changes from top of stash stack

Output -



```
jsk89@JSK MINGW64 ~/git-practical (main)
$ git branch
  feature
* main

jsk89@JSK MINGW64 ~/git-practical (main)
$ git checkout feature
Switched to branch 'feature'

jsk89@JSK MINGW64 ~/git-practical (feature)
$ echo "Feature branch work" >> file.txt

jsk89@JSK MINGW64 ~/git-practical (feature)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

jsk89@JSK MINGW64 ~/git-practical (feature)
$ git commit -m "Added feature"
[feature a569665] Added feature
 1 file changed, 1 insertion(+)

jsk89@JSK MINGW64 ~/git-practical (feature)
$ git checkout master
error: pathspec 'master' did not match any file(s) known to git

jsk89@JSK MINGW64 ~/git-practical (feature)
$ git checkout main
Switched to branch 'main'

jsk89@JSK MINGW64 ~/git-practical (main)
$ git merge feature
Updating f6f4867..a569665
Fast-forward
 file.txt | 1 +
 1 file changed, 1 insertion(+)

jsk89@JSK MINGW64 ~/git-practical (main)
$ git log
commit a569665f6acbac1e15a1ebf6496f26ea4aa1d9a3 (HEAD -> main, feature)
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:02:57 2026 +0530

    Added feature

commit f6f4867c42ec99bb0a0ce11be22fe539e3007a8e
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:02:11 2026 +0530

    Updated file

commit 828600225fca9ae5ba9a2f34a9777b5997aa5d5a
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:01:35 2026 +0530

    Added first file

jsk89@JSK MINGW64 ~/git-practical (main)
```

```
jsk89@JSK MINGW64 ~/git-practical (main)
$ git reset file.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

jsk89@JSK MINGW64 ~/git-practical (main)
$ git commit -m "Added first file"
[main (root-commit) 8286002] Added first file
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ echo "This is my Git practical" >> file.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ git diff
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
diff --git a/file.txt b/file.txt
index 9f4d96d..c2c82cb 100644
--- a/file.txt
+++ b/file.txt
@@ -1 +1,2 @@
 Hello Git
+This is my Git practical

jsk89@JSK MINGW64 ~/git-practical (main)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

jsk89@JSK MINGW64 ~/git-practical (main)
$ git diff --staged
diff --git a/file.txt b/file.txt
index 9f4d96d..c2c82cb 100644
--- a/file.txt
+++ b/file.txt
@@ -1 +1,2 @@
 Hello Git
+This is my Git practical

jsk89@JSK MINGW64 ~/git-practical (main)
$ git commit -m "Updated file"
[main f6f4867] Updated file
 1 file changed, 1 insertion(+)

jsk89@JSK MINGW64 ~/git-practical (main)
$ git branch
* main

jsk89@JSK MINGW64 ~/git-practical (main)
$ git branch feature

jsk89@JSK MINGW64 ~/git-practical (main)
$ git branch
  feature
```

```
    Added first file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git log --oneline
a569665 (HEAD -> main, feature) Added feature
f6f4867 Updated file
8286002 Added first file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git log master..feature
fatal: ambiguous argument 'master..feature': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

jsk89@JSK MINGW64 ~/git-practical (main)
$ git log main..feature

jsk89@JSK MINGW64 ~/git-practical (main)
$ git log --follow file.txt
commit a569665f6acbac1e15a1ebf6496f26ea4aa1d9a3 (HEAD -> main, feature)
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:02:57 2026 +0530

    Added feature

commit f6f4867c42ec99bb0a0ce11be22fe539e3007a8e
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:02:11 2026 +0530

    Updated file

commit 828600225fca9ae5ba9a2f34a9777b5997aa5d5a
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:01:35 2026 +0530

    Added first file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git diff main...feature

jsk89@JSK MINGW64 ~/git-practical (main)
$ git log --oneline
a569665 (HEAD -> main, feature) Added feature
f6f4867 Updated file
8286002 Added first file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git show 8286002
commit 828600225fca9ae5ba9a2f34a9777b5997aa5d5a
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:01:35 2026 +0530

    Added first file

diff --git a/file.txt b/file.txt
new file mode 100644
index 0000000..9f4d96d
```

```
index 0000000..9f4d96d
--- /dev/null
+++ b/file.txt
@@ -0,0 +1 @@
+Hello Git

jsk89@JSK MINGW64 ~/git-practical (main)
$ git mv file.txt newfile.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ git commit -m "Renamed file"
[main 2502af4] Renamed file
 1 file changed, 0 insertions(+), 0 deletions(-)
 rename file.txt => newfile.txt (100%)

jsk89@JSK MINGW64 ~/git-practical (main)
$ git log --stat -M
commit 2502af476453a4ccd860b52c7ddb2499109208a0 (HEAD -> main)
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:06:40 2026 +0530

    Renamed file

 file.txt => newfile.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)

commit a569665f6acbac1e15a1ebf6496f26ea4aa1d9a3 (feature)
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:02:57 2026 +0530

    Added feature

 file.txt | 1 +
 1 file changed, 1 insertion(+)

commit f6f4867c42ec99bb0a0ce11be22fe539e3007a8e
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:02:11 2026 +0530

    Updated file

 file.txt | 1 +
 1 file changed, 1 insertion(+)

commit 828600225fca9ae5ba9a2f34a9777b5997aa5d5a
Author: Jay-2007-coder <jaytalekar82@gmail.com>
Date:   Sat Jul 18 16:01:35 2026 +0530

    Added first file

 file.txt | 1 +
 1 file changed, 1 insertion(+)

jsk89@JSK MINGW64 ~/git-practical (main)
$ echo "Temporary file" > temp.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ git add temp.txt
```

```
$ echo "Temporary file" > temp.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ git add temp.txt
warning: in the working copy of 'temp.txt', LF will be replaced by CRLF the next time Git touches it

jsk89@JSK MINGW64 ~/git-practical (main)
$ git commit -m "Added temporary file"
[main 643b31a] Added temporary file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ git rm temp.txt
rm 'temp.txt'

jsk89@JSK MINGW64 ~/git-practical (main)
$ git commit -m "Deleted temporary file"
[main c503d5c] Deleted temporary file
 1 file changed, 1 deletion(-)
 delete mode 100644 temp.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ echo "Work in progress" >> newfile.txt

jsk89@JSK MINGW64 ~/git-practical (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   newfile.txt

no changes added to commit (use "git add" and/or "git commit -a")

jsk89@JSK MINGW64 ~/git-practical (main)
$ git stash
warning: in the working copy of 'newfile.txt', LF will be replaced by CRLF the next time Git touches it
Saved working directory and index state WIP on main: c503d5c Deleted temporary file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git stash list
stash@{0}: WIP on main: c503d5c Deleted temporary file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git stash pop
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   newfile.txt

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (741cef0cbdfc84681f44e1fafda0ab8d4fcbe331)

jsk89@JSK MINGW64 ~/git-practical (main)
$ git stash
Saved working directory and index state WIP on main: c503d5c Deleted temporary file
```

```
jsk89@JSK MINGW64 ~/git-practical (main)
$ git stash
Saved working directory and index state WIP on main: c503d5c Deleted temporary file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git stash list
stash@{0}: WIP on main: c503d5c Deleted temporary file

jsk89@JSK MINGW64 ~/git-practical (main)
$ git stash drop
Dropped refs/stash@{0} (a674d927a65e1cbecb98301c66c511839c38df40)

jsk89@JSK MINGW64 ~/git-practical (main)
$ git checkout -b development
Switched to a new branch 'development'

jsk89@JSK MINGW64 ~/git-practical (development)
$ echo "Development work" >> newfile.txt

jsk89@JSK MINGW64 ~/git-practical (development)
$ git add newfile.txt
warning: in the working copy of 'newfile.txt', LF will be replaced by CRLF the next time Git touches it

jsk89@JSK MINGW64 ~/git-practical (development)
$ git commit -m "Development changes"
[development 71fcf05] Development changes
 1 file changed, 1 insertion(+)

jsk89@JSK MINGW64 ~/git-practical (development)
$ git rebase main
Current branch development is up to date.

jsk89@JSK MINGW64 ~/git-practical (development)
$ git log --oneline
71fcf05 (HEAD -> development) Development changes
c503d5c (main) Deleted temporary file
643b31a Added temporary file
2502af4 Renamed file
a569665 (feature) Added feature
f6f4867 Updated file
8286002 Added first file

jsk89@JSK MINGW64 ~/git-practical (development)
$ git reset --hard 8286002
HEAD is now at 8286002 Added first file

jsk89@JSK MINGW64 ~/git-practical (development)
$ git remote add origin https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git

jsk89@JSK MINGW64 ~/git-practical (development)
$ git remote -v
origin  https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git (fetch)
origin  https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git (push)

jsk89@JSK MINGW64 ~/git-practical (development)
$ git push -u origin main
Enumerating objects: 15, done.
```

```
$ git remote -v
origin  https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git (fetch)
origin  https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git (push)

jsk89@JSK MINGW64 ~/git-practical (development)
$ git push -u origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (15/15), 1.29 KiB | 1.29 MiB/s, done.
Total 15 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

jsk89@JSK MINGW64 ~/git-practical (development)
$ git fetch origin

jsk89@JSK MINGW64 ~/git-practical (development)
$ git merge origin/main
Updating 8286002..c503d5c
Fast-forward
 file.txt    | 1 -
 newfile.txt | 3 +++
 2 files changed, 3 insertions(+), 1 deletion(-)
 delete mode 100644 file.txt
 create mode 100644 newfile.txt

jsk89@JSK MINGW64 ~/git-practical (development)
$ git pull
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

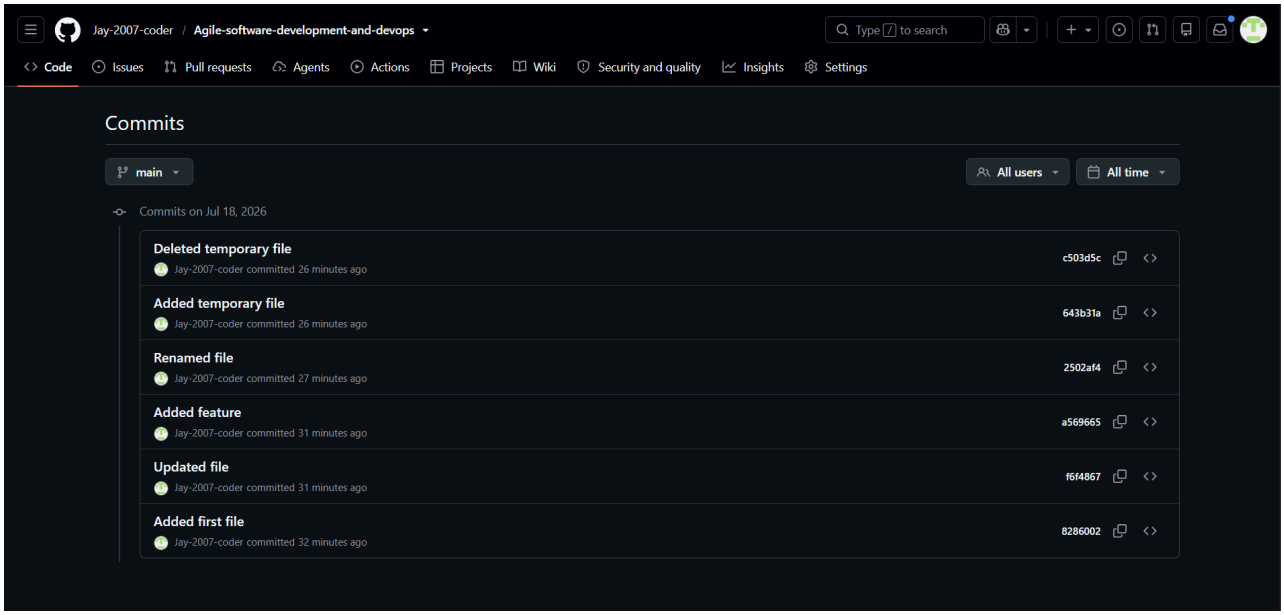
    git branch --set-upstream-to=origin/<branch> development


jsk89@JSK MINGW64 ~/git-practical (development)
$ cd ..

jsk89@JSK MINGW64 ~
$ git clone https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git
Cloning into 'Agile-software-development-and-devops'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 15 (delta 1), reused 15 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), done.
Resolving deltas: 100% (1/1), done.

jsk89@JSK MINGW64 ~
$
```

Github Repo -





Conclusion:

Q1. What is a Version Control System ?

Ans - A **Version Control System (VCS)** is a software tool used to track and manage changes made to source code and files over time. It allows developers to maintain different versions of a project, restore previous versions, track modifications, and collaborate with multiple developers efficiently.

Q2. Which is the difference between GIT and GIThub?

Ans - **Git** is a distributed version control system used to track changes in source code locally. It allows developers to create commits, branches, merge changes, and maintain the complete history of a project.**GitHub** is a cloud-based platform used to host and manage Git repositories online. It provides features for collaboration, code sharing, pull requests, issue tracking, and project management.**In simple terms:** Git is a tool used for version control, while GitHub is an online platform used to store and collaborate on Git repositories.

Q3. How to retrieve an entire repository from a hosted location via URL?

Ans - To retrieve or download an entire repository from a hosted location such as GitHub, the git clone command is used.

Syntax:

git clone [repository URL]

Command used in the practical:

git clone https://github.com/Jay-2007-coder/Agile-software-development-and-devops.git

This command creates a local copy of the complete GitHub repository, including its files and commit history, on the computer.

Q4. How to change an existing file path?

Ans - The git mv command is used to move a file to another location or rename an existing file while tracking the change in Git.

Syntax:

git mv [existing-path] [new-path]

Example:

git mv file.txt newfolder/file.txt

After changing the file path, the changes can be committed using:

git commit -m "Changed file path"

This command moves the file from its existing path to the new path and stages the change for the next commit.